Method I

–1	1	13	32	20	
	0	–1	–12	–20	
–2	1	12	20	0	(remainder)
	0	–2	–20		
	1	10	0	(remainder)	

$p(x) = (x+1)(x+2)(x+10)$

Hence,

$x^3 + 13x^2 + 32x + 20$
$= (x+1)(x+2)(x+10)$

Method II

–1	1	13	32	20
	0	–1	–12	–20
	1	12	20	0 (remainder)

Then $p(x) = (x+1)(x^2+12x+20)$

Now $x^2 + 12x + 20 = x^2 + 10x + 2x + 20$
$= x(x+10) + 2(x+10)$
$= (x+2)(x+10)$

Hence, $x^3 + 13x^2 + 32x + 20$
$= (x+1)(x+2)(x+10)$

Example 3.40

Factorise $x^3 - 5x^2 - 2x + 24$

Solution

Let $p(x) = x^3 - 5x^2 - 2x + 24$

When $x = 1, p(1) = 1 - 5 - 2 + 24 = 18 \neq 0$ $(x-1)$ is not a factor.

When $x = -1$, $p(-1) = -1 - 5 + 2 + 24 = 20 \neq 0$ $(x+1)$ is not a factor.

Therefore, we have to search for different values of x by trial and error method.

When $x = 2$

$p(2) = 2^3 - 5(2)^2 - 2(2) + 24$
$= 8 - 20 - 4 + 24$
$= 8 \neq 0$ Hence, $(x-2)$ is not a factor

When $x = -2$

$p(-2) = (-2)^3 - 5(-2)^2 - 2(-2) + 24$
$= -8 - 20 + 4 + 24$

$p(-2) = 0$

Hence, $(x+2)$ is a factor

–2	1	–5	–2	24	
	0	–2	+14	–24	
3	1	–7	12	0	(remainder)
	0	3	–12		
	1	–4	0	(remainder)	

Thus, $(x+2)(x-3)(x-4)$ are the factors.

Therefore, $x^3 - 5x^2 - 2x^2 + 24 = (x+2)(x-3)(x-4)$

Note

Check whether 3 is a zero of $x^2 - 7x + 12$. If it is not, then check for –3 or 4 or –4 and so on.

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